Result Interpretation based on paper:

AMULTI-NOISEMULTI-CHANNELANCSYSTEMUSINGRELATIVETRANSFER MATRIX-BASEDAPPROACH

- Simulation Scenario:

“We conduct an ANC system with J = 2 noise sources at [0.6, 0.8, 1] m and [2.17,1.15,0.05] m, K = 2 secondary loudspeakers at [0,±0.3,0] m, and M = 8 monitoring microphones at [±0.15,±0.15,−0.15] m, [0,±0.212,0.15] m and [±0.212,0,0.15] m, in a simulated room of size [8,7,4] m and T60 = 500ms. The impulse responses are obtained using the image source method. For the tuning stage, we place V = 6 of microphones at [0.05,±0.10,0] m, [−0.05,±0.10,0] m and [0,±0.10,0] m to obtain the necessary ReTM mapping. The last pair of microphones at [0, ±0.10,0] are used to examine the performance of ANC algorithms. We use two broadband factory noise signals of length 100 s from the NOISEX-92 dataset with 8 kHz sampling rate.”

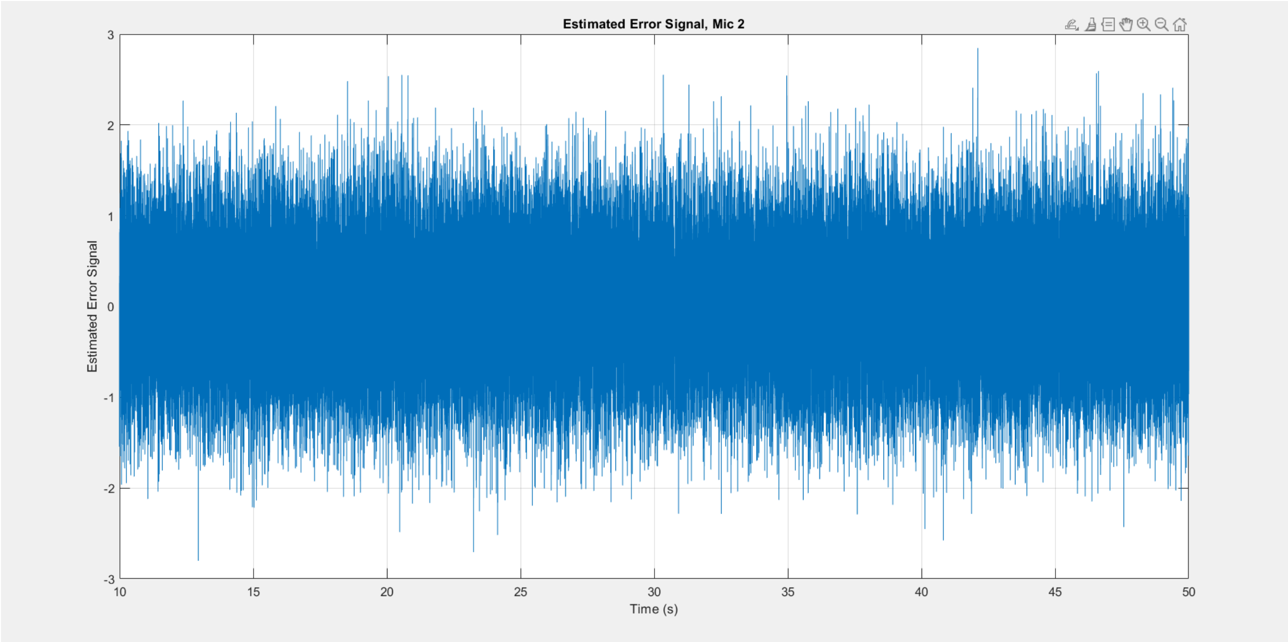
- from the paper

- 2 noise sources: [0.6, 0.8, 1] , [2.17,1.15,0.05]
- 2 Secondary Loudspeakers: [0,±0.3,0]
- 8 monitoring mics: [±0.15,±0.15,−0.15], [0,±0.212,0.15], [±0.212,0,0.15]
- Room Size: [8,7,4]
- 4 Virtual Error mics: [0.05,±0.10,0], [−0.05,±0.10,0]
- 2 Evaluation mics: [0,±0.10,0]
- Sampling Rate: 8kHz
- Duration of the source signal: 60s

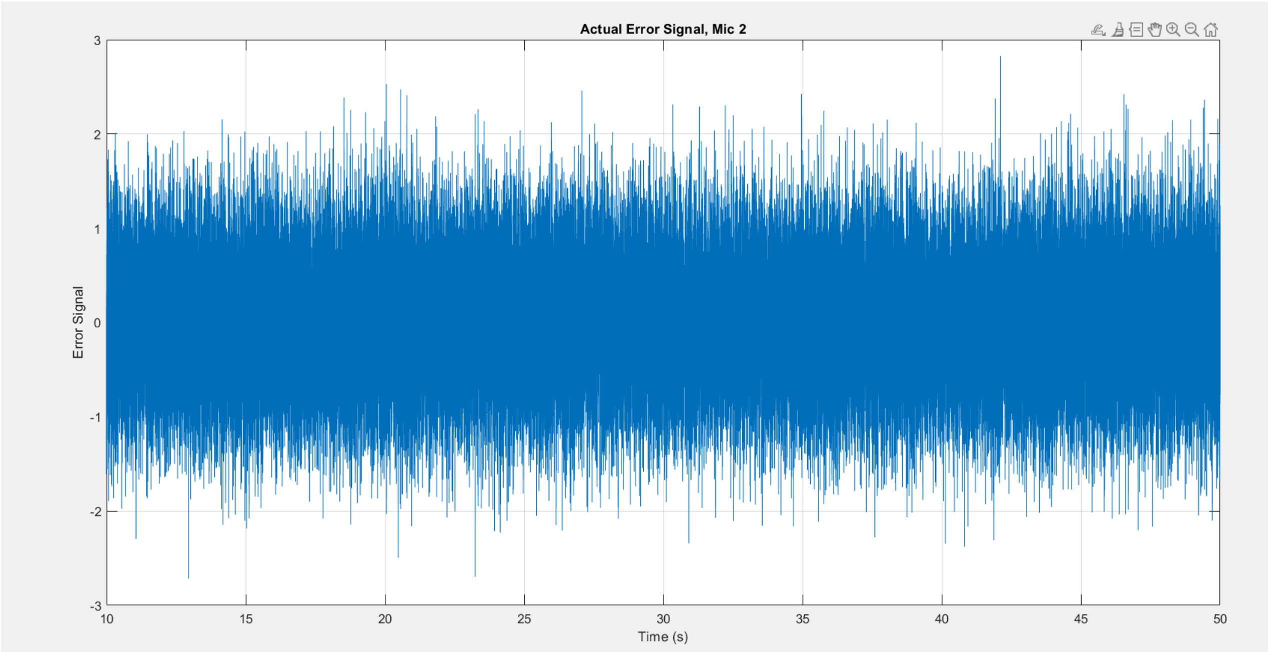
- Results:

1. Tuning Stage:

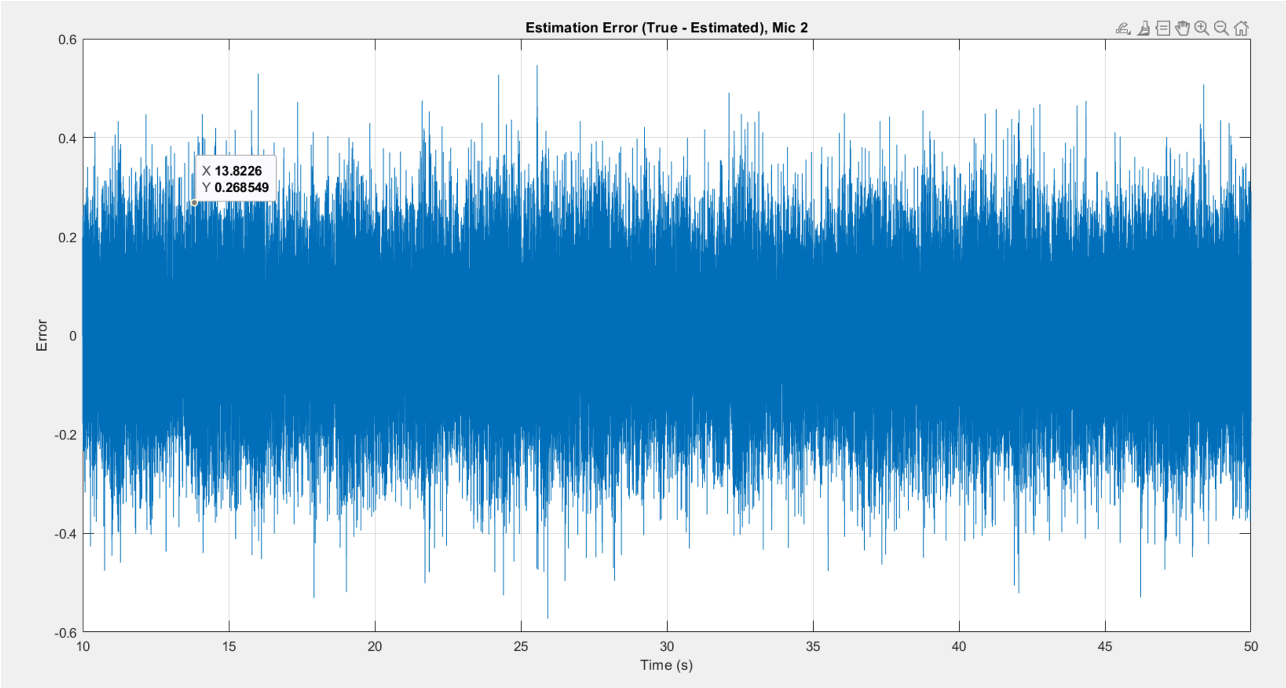
Used 2 factory noise audio files as primary source and used a random noise between 20-600Hz frequency. Then using the ReTM calculation function to find ReTM. Then see the error plot of the actual error microphone signal and estimated one.



Estimated Error mic Signal



Actual Error mic signal



Error of the estimation

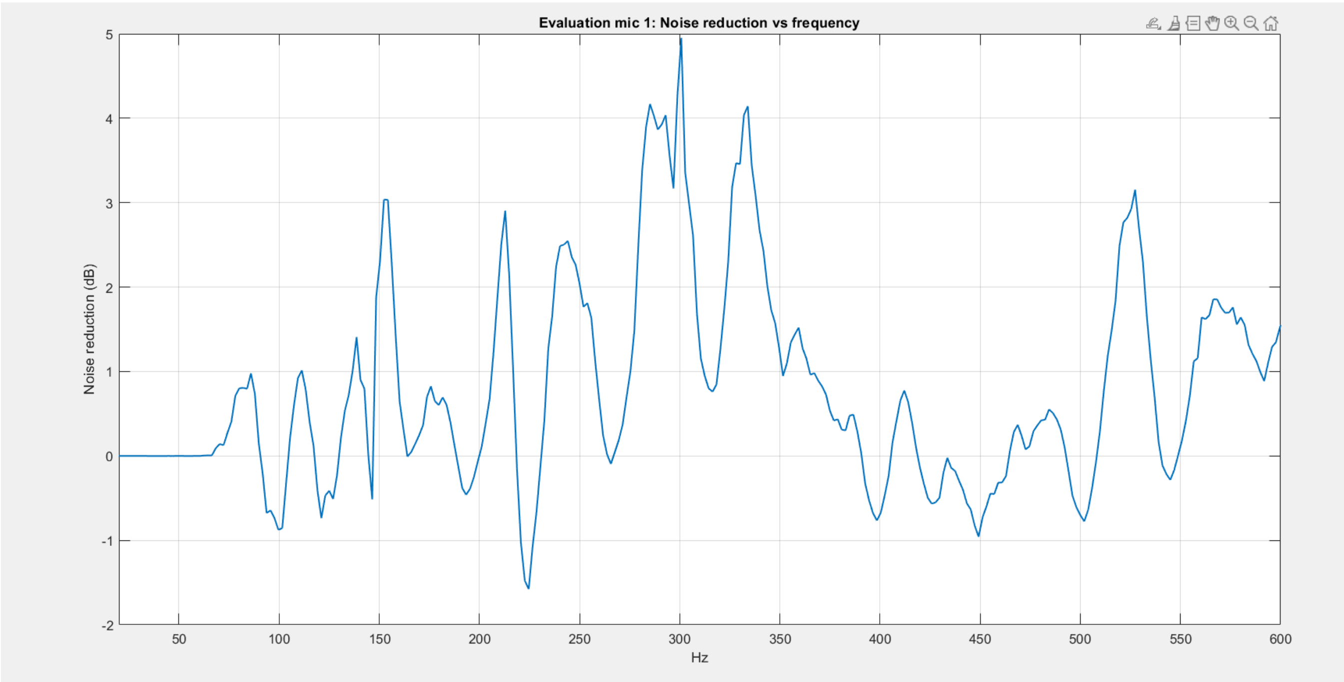
Conclusion:

Error between the two error mic signals are less than 0. So, we can think both actual and estimated error mic signals are the same for this scenario with ReTM.

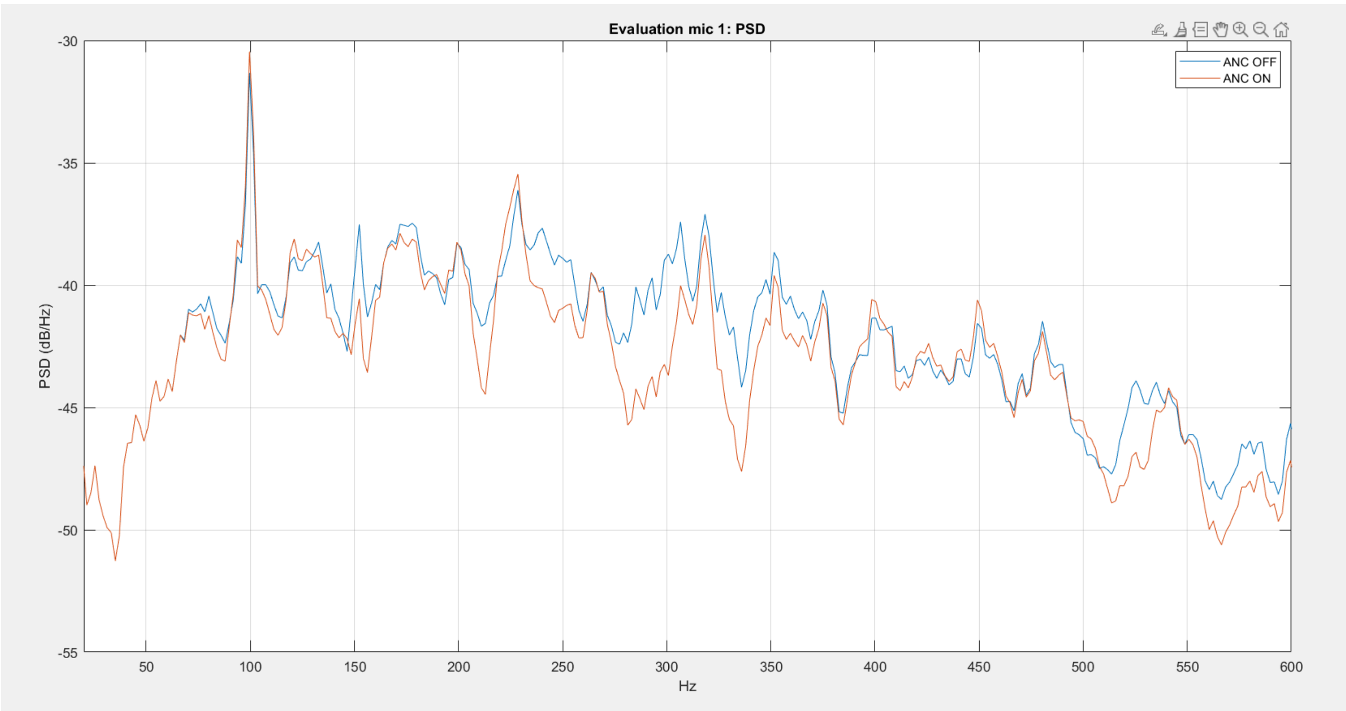
2. Control Stage

Used the same primary source audio files to generate the primary signal. Then cancel out the noise frequencies between 20-600Hz using the secondary microphones. Used ReTM estimation matrix calculated in tuning stage to this stage. Get a 20s - 40s audio block from original audio to generate the virtual error signal from the monitoring signal. Then implement the ANC loop on STFT domain and plot the evaluation microphone signals when ANC off and ANC on states.

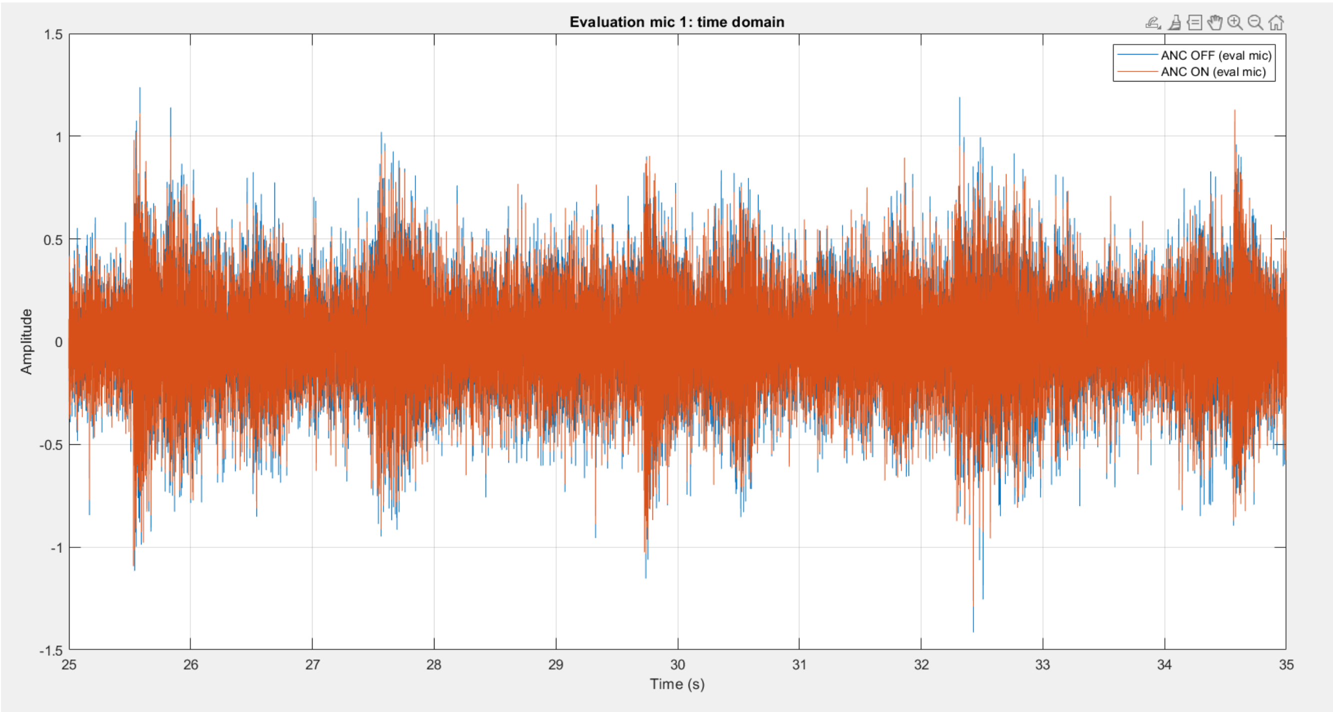
Plot 1: Noise Reduction Vs Frequency:



Plot 2: PSD at Error microphones



Plot 3: Actual Time domain error signals.



Conclusion:

From this simulation I could achieve nearly 5dB maximum noise cancellation according to the plots. Also, paper shows they achieved that amount of noise cancellation.